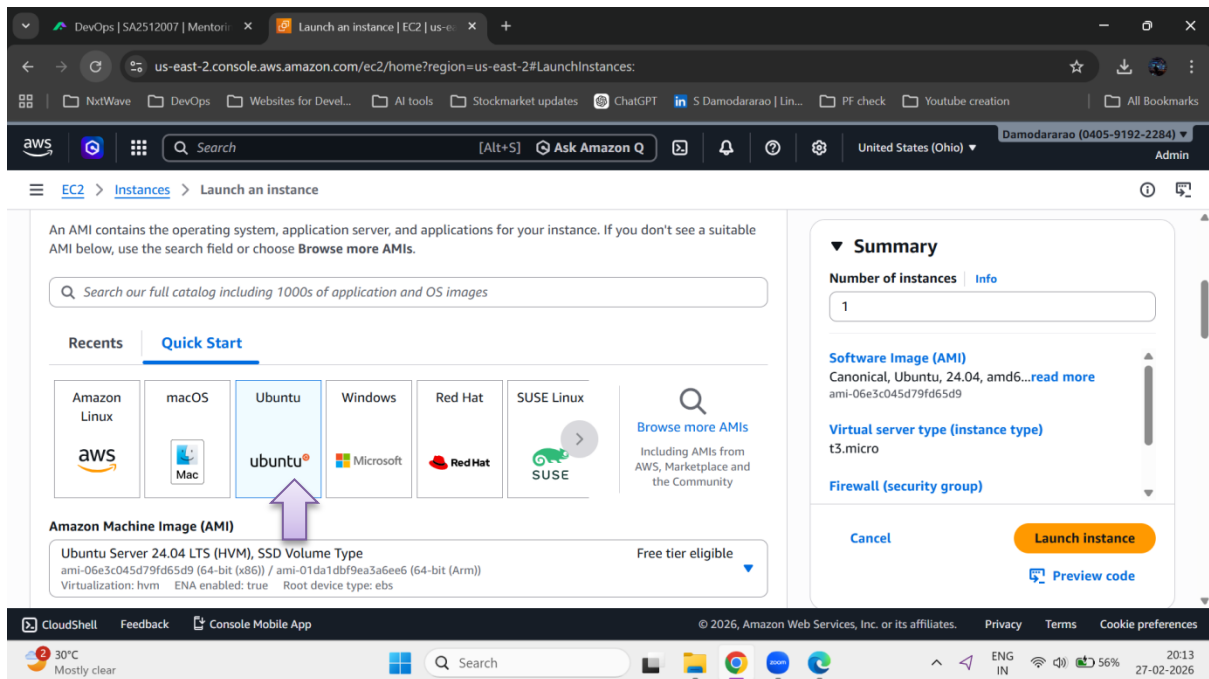


Assignment – 4

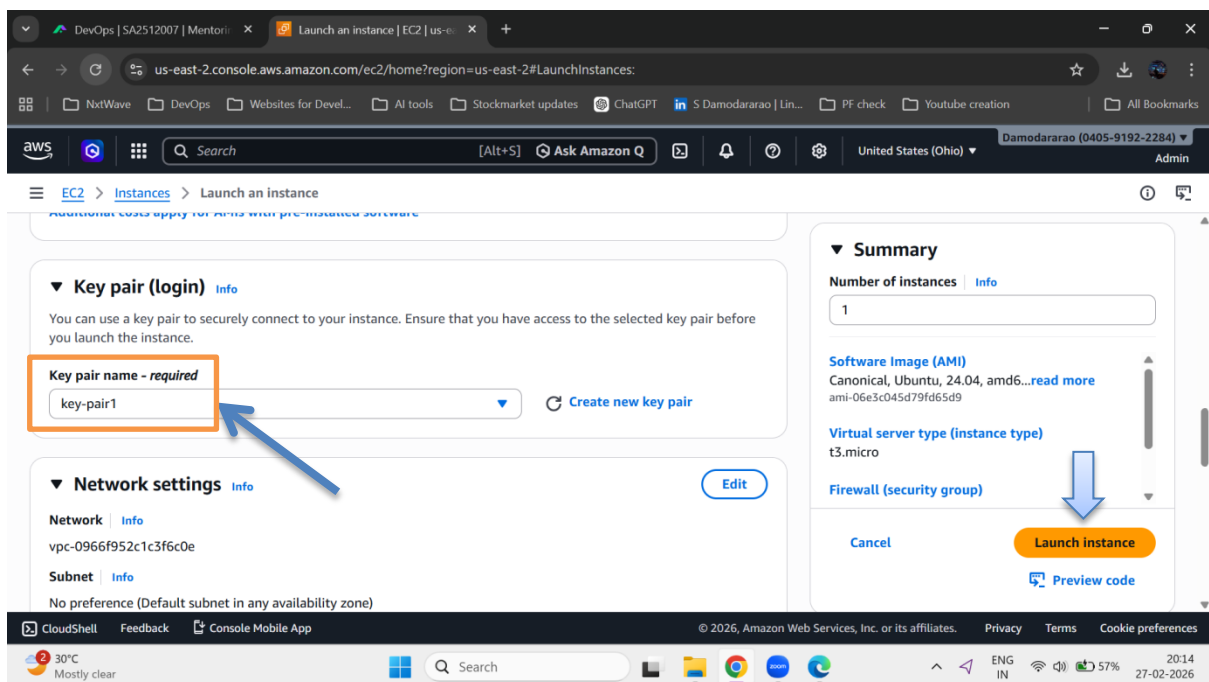
Connect to the AWS EC2 Ubuntu Instance and Update default packages, install JDK, Maven, Git, and validate the versions

→ Go to EC2 Dashboard & Click Launch Instance

- **Name:** Server-1
- **AMI:** Ubuntu Server 24.04 LTS & select **Instance Type:** t3.micro



➤ Key Pair: Create or Select key pair and Launch instance



➤ Select your running Ubuntu instance

The screenshot shows the AWS Management Console for the EC2 Instances page. The left sidebar contains navigation links for EC2, Dashboard, AWS Global View, Events, and Instances. The main content area displays a table of instances. The instance 'server-1' is highlighted with a red box. The table columns are Name, Instance ID, Instance state, Instance type, Status check, and Alarm status. The instance 'server-1' has an Instance ID of 'i-0834e0c983bd2f70e' and is in the 'Running' state. The instance type is 't3.micro' and the status check is 'Initializing'.

✓ Click Connect & Click EC2 Instance Connect

The screenshot shows the AWS Management Console for the 'Connect to instance' page. The left sidebar contains navigation links for EC2, Instances, and Connect to instance. The main content area displays the 'EC2 Instance Connect' tab, which is highlighted with a red box. The instance ID 'i-0834e0c983bd2f70e' is shown. The connection type is set to 'Public IPv4 address' with the address '3.17.184.16'. The username is set to 'ubuntu'.

➤ Click on Connect

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#ConnectToInstance:instanceId=i-0834e0c983bd2f70e

EC2 > Instances > i-0834e0c983bd2f70e > Connect to instance

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

Public IPv4 address
3.17.184.16

IPv6 address
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.
ubuntu

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel **Connect**

➤ Connecting Successfully

us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-2&connType=standard&instanceId=i-0834e0c983bd2f70e&osUser=...

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.
To check for new updates run: `sudo apt update`

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in `/usr/share/doc/*/copyright`.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "`sudo <command>`".
See "`man sudo_root`" for details.

ubuntu@ip-172-31-11-3:~\$

i-0834e0c983bd2f70e (server-1)
PublicIPs: 3.17.184.16 PrivateIPs: 172.31.11.3

➤ Update Packages. Command: `sudo apt upgrade -y`

```
ubuntu@ip-172-31-11-3:~$ sudo apt upgrade -y
Hit:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1494 kB]
Get:8 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:9 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:10 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:11 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:12 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:13 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:14 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1800 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [240 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [9932 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [961 kB]
```

i-0834e0c983bd2f70e (server-1)

PublicIPs: 3.17.184.16 PrivateIPs: 172.31.11.3

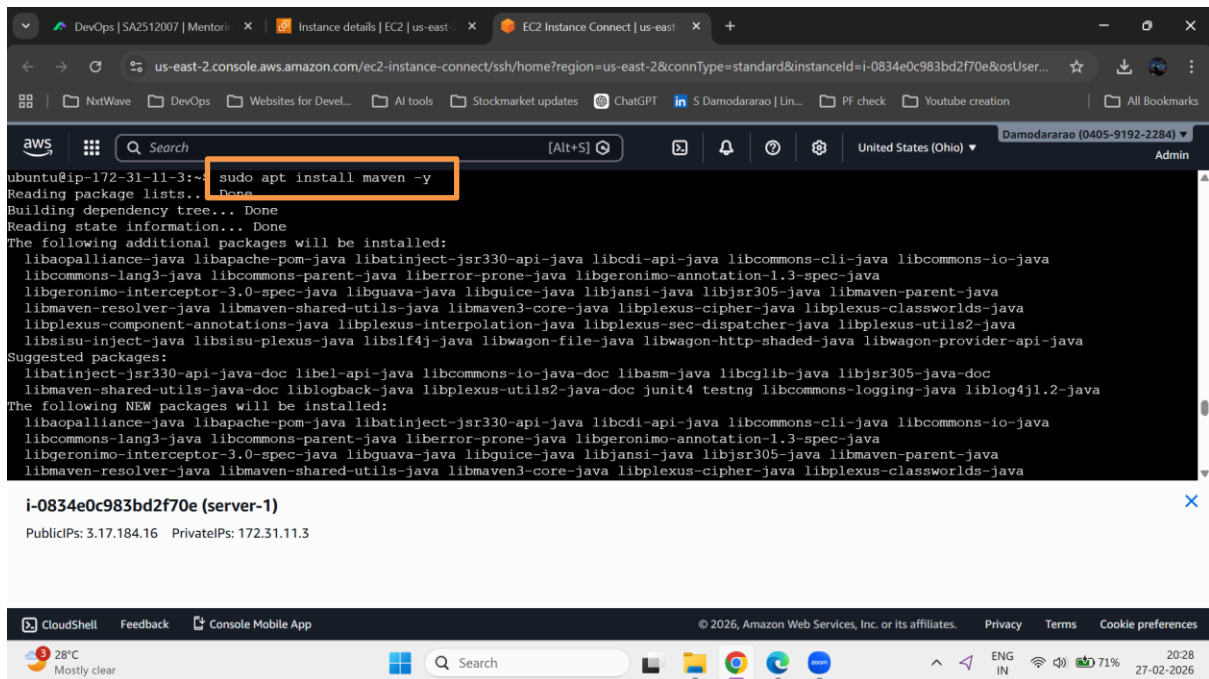
➤ Install Java. Command : `sudo apt install openjdk-21-jdk -y`

```
ubuntu@ip-172-31-11-3:~$ sudo apt install openjdk-21-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcolor2 libepoxy0 libgtk-3-0t64 libgtk-3-bin libgtk-3-common libwayland-cursor0 libwayland-egl1 openjdk-21-jdk-headless
  openjdk-21-jre openjdk-21-jre-headless
Suggested packages:
  colord gvfs openjdk-21-demo openjdk-21-source visualvm libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei
  | fonts-wqy-zenhei fonts-indic
The following NEW packages will be installed:
  libcolor2 libepoxy0 libgtk-3-0t64 libgtk-3-bin libgtk-3-common libwayland-cursor0 libwayland-egl1 openjdk-21-jdk
  openjdk-21-jdk-headless openjdk-21-jre openjdk-21-jre-headless
0 upgraded, 11 newly installed, 0 to remove and 97 not upgraded.
Need to get 136 MB of archives.
After this operation, 318 MB of additional disk space will be used.
Get:1 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libcolor2 amd64 1.4.7-1build2 [149 kB]
Get:2 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libepoxy0 amd64 1.5.10-1build1 [220 kB]
Get:3 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libwayland-cursor0 amd64 1.22.0-2.1build1 [10.4 kB]
```

i-0834e0c983bd2f70e (server-1)

PublicIPs: 3.17.184.16 PrivateIPs: 172.31.11.3

➤ Install Maven. Command: `sudo apt install maven -y`

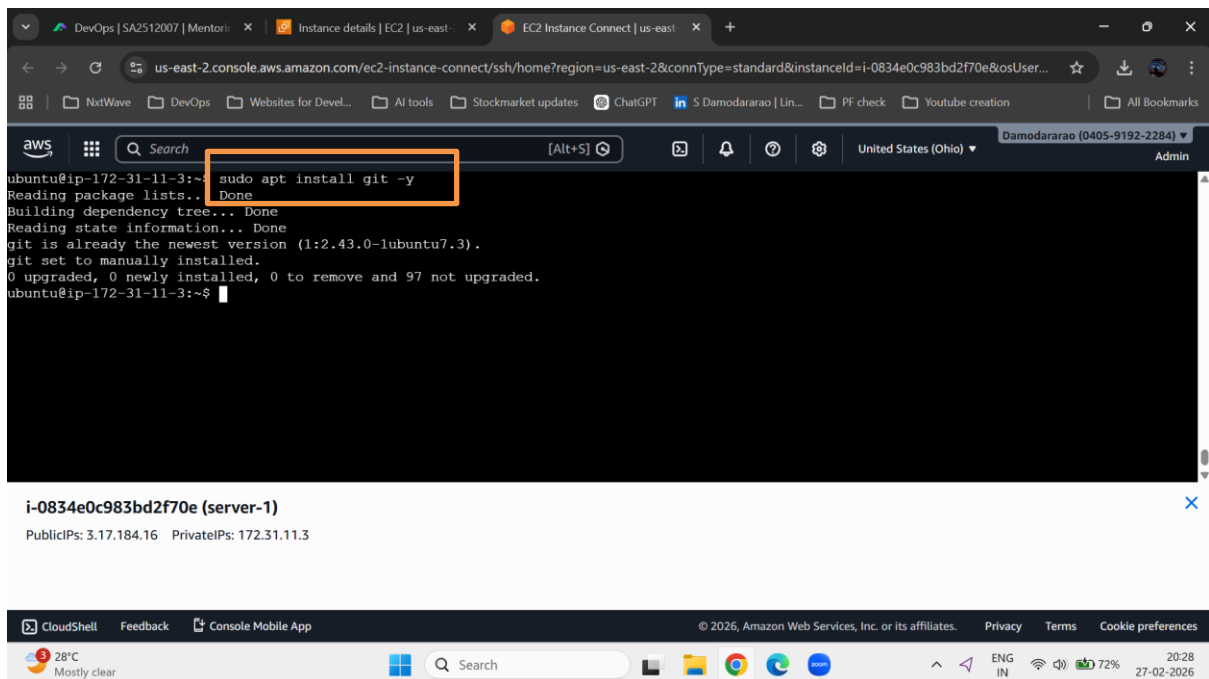


The screenshot shows a terminal window in the AWS CloudShell environment. The command `sudo apt install maven -y` has been executed. The terminal output shows the package lists being read, the dependency tree being built, and the state information being read. It then lists the additional packages that will be installed, including `libaopalliance-java`, `libapache-pom-java`, `libatinject-jsr330-api-java`, `libcdi-api-java`, `libcommons-cli-java`, `libcommons-io-java`, `libcommons-lang3-java`, `libcommons-parent-java`, `liberror-prone-java`, `libgeronimo-annotation-1.3-spec-java`, `libgeronimo-interceptor-3.0-spec-java`, `libguava-java`, `libguice-java`, `libjansi-java`, `libjsr305-java`, `libmaven-parent-java`, `libmaven-resolver-java`, `libmaven-shared-utils-java`, `libmaven3-core-java`, `libplexus-cipher-java`, `libplexus-classworlds-java`, `libplexus-component-annotations-java`, `libplexus-interpolation-java`, `libplexus-sec-dispatcher-java`, `libplexus-utils2-java`, `libsisu-inject-java`, `libsisu-plexus-java`, `libslf4j-java`, `libwagon-file-java`, `libwagon-http-shaded-java`, and `libwagon-provider-api-java`. It also lists suggested packages like `libatinject-jsr330-api-java-doc`, `libel-api-java`, `libcommons-io-java-doc`, `libasm-java`, `libcglib-java`, `libjsr305-java-doc`, `libmaven-shared-utils-java-doc`, `liblogback-java`, `libplexus-utils2-java-doc`, `junit4`, `testng`, `libcommons-logging-java`, and `liblog4j1.2-java`. The terminal output ends with the message "The following NEW packages will be installed:" followed by the same list of packages. The terminal window is titled "i-0834e0c983bd2f70e (server-1)" and shows the public IP as 3.17.184.16 and the private IP as 172.31.11.3. The bottom of the screenshot shows the AWS CloudShell interface with the "Feedback" button and the "Console Mobile App" link. The system status bar at the bottom shows the temperature as 28°C, the time as 20:28, and the date as 27-02-2026.

```
ubuntu@ip-172-31-11-3:~$ sudo apt install maven -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libaopalliance-java libapache-pom-java libatinject-jsr330-api-java libcdi-api-java libcommons-cli-java libcommons-io-java
  libcommons-lang3-java libcommons-parent-java liberror-prone-java libgeronimo-annotation-1.3-spec-java
  libgeronimo-interceptor-3.0-spec-java libguava-java libguice-java libjansi-java libjsr305-java libmaven-parent-java
  libmaven-resolver-java libmaven-shared-utils-java libmaven3-core-java libplexus-cipher-java libplexus-classworlds-java
  libplexus-component-annotations-java libplexus-interpolation-java libplexus-sec-dispatcher-java libplexus-utils2-java
  libsisu-inject-java libsisu-plexus-java libslf4j-java libwagon-file-java libwagon-http-shaded-java libwagon-provider-api-java
Suggested packages:
  libatinject-jsr330-api-java-doc libel-api-java libcommons-io-java-doc libasm-java libcglib-java libjsr305-java-doc
  libmaven-shared-utils-java-doc liblogback-java libplexus-utils2-java-doc junit4 testng libcommons-logging-java liblog4j1.2-java
The following NEW packages will be installed:
  libaopalliance-java libapache-pom-java libatinject-jsr330-api-java libcdi-api-java libcommons-cli-java libcommons-io-java
  libcommons-lang3-java libcommons-parent-java liberror-prone-java libgeronimo-annotation-1.3-spec-java
  libgeronimo-interceptor-3.0-spec-java libguava-java libguice-java libjansi-java libjsr305-java libmaven-parent-java
  libmaven-resolver-java libmaven-shared-utils-java libmaven3-core-java libplexus-cipher-java libplexus-classworlds-java
```

i-0834e0c983bd2f70e (server-1)
PublicIPs: 3.17.184.16 PrivateIPs: 172.31.11.3

➤ Install Git. Command: `sudo apt install git -y`



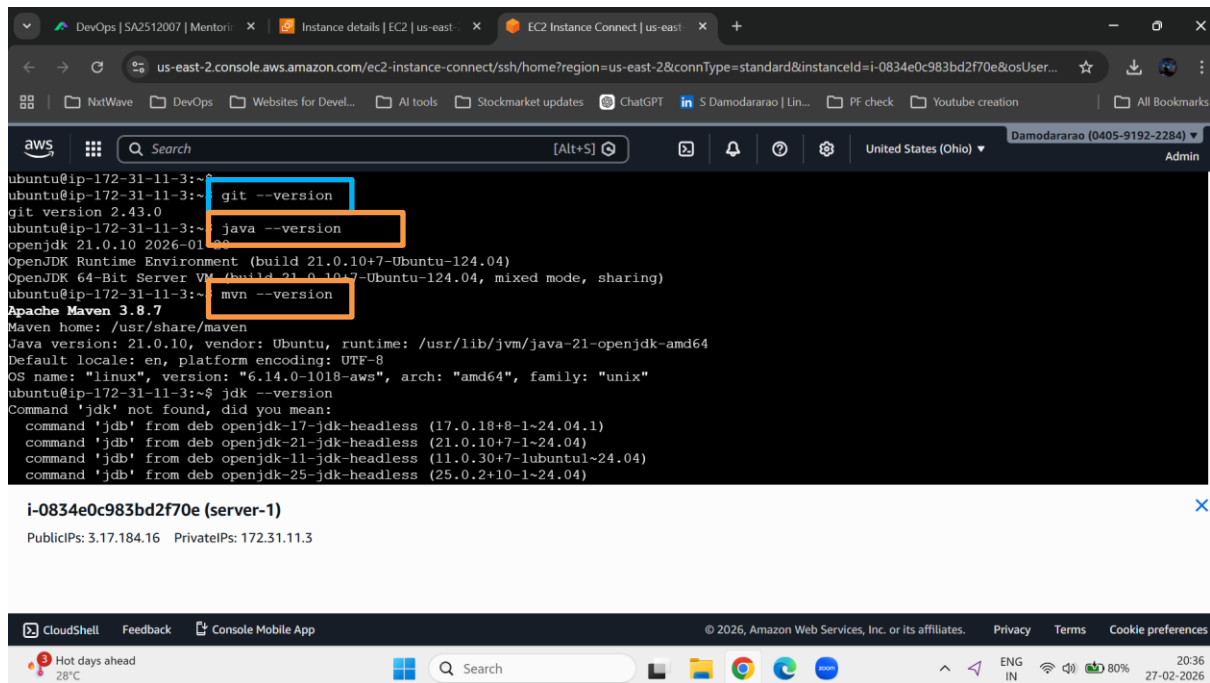
The screenshot shows a terminal window in the AWS CloudShell environment. The command `sudo apt install git -y` has been executed. The terminal output shows the package lists being read, the dependency tree being built, and the state information being read. It then shows that git is already the newest version (1:2.43.0-1ubuntu7.3) and is set to manually installed. The terminal output ends with the message "0 upgraded, 0 newly installed, 0 to remove and 97 not upgraded." The terminal window is titled "i-0834e0c983bd2f70e (server-1)" and shows the public IP as 3.17.184.16 and the private IP as 172.31.11.3. The bottom of the screenshot shows the AWS CloudShell interface with the "Feedback" button and the "Console Mobile App" link. The system status bar at the bottom shows the temperature as 28°C, the time as 20:28, and the date as 27-02-2026.

```
ubuntu@ip-172-31-11-3:~$ sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 97 not upgraded.
ubuntu@ip-172-31-11-3:~$
```

i-0834e0c983bd2f70e (server-1)
PublicIPs: 3.17.184.16 PrivateIPs: 172.31.11.3

Check for conformation with version commands

1) Java --version 2) mvn --version 3) git --version



The screenshot shows a terminal window within the AWS CloudShell interface. The terminal output displays the results of three version commands: `git --version`, `java --version`, and `mvn --version`. The `git` command returns version 2.43.0. The `java` command returns information about the OpenJDK Runtime Environment (build 21.0.10+7-Ubuntu-124.04) and the OpenJDK 64-Bit Server VM (build 21.0.10+7-Ubuntu-124.04, mixed mode, sharing). The `mvn` command returns Apache Maven 3.8.7. The terminal also shows the Maven home directory, Java version, vendor, runtime, default locale, platform encoding, and OS name. The AWS CloudShell interface includes a search bar, navigation icons, and a status bar at the bottom showing the instance ID, public and private IP addresses, and the time.

```
ubuntu@ip-172-31-11-3:~$ git --version
git version 2.43.0
ubuntu@ip-172-31-11-3:~$ java --version
openjdk 21.0.10 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+7-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 21.0.10+7-Ubuntu-124.04, mixed mode, sharing)
ubuntu@ip-172-31-11-3:~$ mvn --version
Apache Maven 3.8.7
Maven home: /usr/share/maven
Java version: 21.0.10, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.14.0-1018-aws", arch: "amd64", family: "unix"
ubuntu@ip-172-31-11-3:~$ jdk --version
Command 'jdk' not found, did you mean:
  command 'jdb' from deb openjdk-17-jdk-headless (17.0.18+8-1~24.04.1)
  command 'jdb' from deb openjdk-21-jdk-headless (21.0.10+7-1~24.04)
  command 'jdb' from deb openjdk-11-jdk-headless (11.0.30+7-1ubuntu1~24.04)
  command 'jdb' from deb openjdk-25-jdk-headless (25.0.2+10-1~24.04)
```

i-0834e0c983bd2f70e (server-1)
PublicIPs: 3.17.184.16 PrivateIPs: 172.31.11.3

Connected to the AWS EC2 Ubuntu Instance and Update default packages, install JDK, Maven, Git, and validated the versions